Bitcoin Market Sentiment and Trader Performance Analysis

1. Introduction -

This project analyzes the relationship between market sentiment and trader performance using two datasets: the Bitcoin Fear & Greed Index and historical trading data from Hyperliquid.

By combining these datasets, the analysis aims to determine whether trading behavior and profitability vary across different sentiment conditions such as Fear, Greed, Extreme Fear, Extreme Greed, and Neutral market states.

The primary objective is to identify patterns in trading activity, profitability, win rates, trading volume, and position preferences under different market sentiments and derive actionable insights that can support smarter trading decisions.

2. Dataset Overview -

Bitcoin Fear & Greed Index Dataset:

The Bitcoin Fear & Greed Index dataset measures overall market sentiment on a daily basis. The index ranges from 0 to 100 and classifies market conditions into categories such as Extreme Fear, Fear, Neutral, Greed, and Extreme Greed.

Key columns include:

- Date: Date of sentiment measurement
- Value: Numerical Fear & Greed score
- Classification: Sentiment category (Fear, Greed, etc.)

This dataset serves as a representation of the emotional state of the market.

Historical Trader Dataset:

The historical trading dataset contains detailed transaction records from Hyperliquid traders. Each record represents a trading activity and includes information related to position type, trade size, execution price, profit and loss and trading direction.

Important columns include:

- Account: Trader identifier
- Coin: Cryptocurrency traded
- Execution Price: Price at which the trade was executed
- Size USD: Trade value in USD
- Direction: Type of trading action
- Timestamp IST: Date and time of trade execution

- Closed PnL: Realized profit or loss from the trade
- Fee: Trading fee charged

This dataset is used to evaluate trader performance under different market sentiment conditions.

3. Project Objective -

The main objectives of this analysis are:

- Analyze trading activity under different market sentiments.
- Compare trader profitability across Fear, Greed, Neutral, Extreme Fear, and Extreme Greed conditions.
- Evaluate trader win rates under different market sentiments.
- Analyze trading volume and participation across sentiment categories.
- Investigate the performance of long and short trading strategies under varying market conditions.
- Identify coin trading preferences under different market sentiments.
- Generate insights that can help traders improve decision-making and risk management strategies.

4. Data Cleaning and Preprocessing -

Before performing the analysis, both datasets were examined and cleaned to ensure consistency and reliability.

i. Data Inspection:

The datasets were initially explored using functions such as .head() , .shape(), .columns(), and .info() to understand their structure, dimensions, and data types.

ii. Missing Value Check:

Missing values were identified using the .isnull().sum() function.
Results showed that:

- The historical trading dataset contained no missing values.
- The Fear & Greed Index dataset contained no missing values.

Since no missing values were present, no imputation or row removal was required.

iii. Duplicate Record Check:

Duplicate records were checked using the . duplicated (). sum () function.
Results indicated:

- No duplicate records in the historical trading dataset.
- No duplicate records in the Fear & Greed dataset.

iv. Date Format Conversion:

To enable dataset integration, date columns were converted into a consistent datetime format.
The following transformations were performed:

- The date column in the Fear & Greed dataset was converted to datetime format.
- The Timestamp IST column in the historical trading dataset was converted to datetime format.
- A new date column was extracted from the trade timestamp to match the sentiment dataset.

This standardization allowed accurate alignment of trading records with daily market sentiment information.

v. Dataset Merging:

After preprocessing, both datasets were merged using the common date field.
The merge process attached the corresponding market sentiment classification to each trade, allowing every transaction to be analyzed within its respective market sentiment context.
The final merged dataset formed the basis for all subsequent analyses.

5. Methodology –

The analysis was conducted using Exploratory Data Analysis (EDA) techniques to investigate how trader behavior and performance varied under different market sentiment conditions.
The following metrics were evaluated:

I. Trade Activity Analysis:

The total number of trades executed under each sentiment category was calculated to identify periods of increased market participation.

II. Profitability Analysis:

Average Closed Profit and Loss (Closed PnL) was calculated for each sentiment category to determine which market conditions produced the highest profitability per trade.
Total Closed PnL was also computed to measure the overall profitability generated under each sentiment.

III. Win Rate Analysis:

A trade was classified as a winning trade when its Closed PnL was greater than zero.
Win Rate was calculated as:
Win Rate = (Number of Profitable Trades / Total Number of Trades) × 100
This metric was used to evaluate trading success across different market conditions.

IV. Trading Volume Analysis:

Trading volume was measured using the Size USD column.

The total trading volume under each sentiment category was analyzed to determine where traders committed the highest amount of capital.

V. Long vs Short Performance Analysis:

Average profitability was compared between long and short positions under each sentiment category.

This analysis helped identify whether bullish or bearish trading strategies performed better under specific market conditions.

VI. Coin Preference Analysis:

The most frequently traded cryptocurrency under each sentiment category was identified.

This analysis provided insights into changes in trader preferences during different market environments.

6. Visualization –

Bar charts were created to visually compare:

- Number of trades by sentiment
- Average profit by sentiment
- Win rate by sentiment
- Trading volume by sentiment

These visualizations helped simplify interpretation and communicate findings more effectively.

7. Results and Findings –

I. Trade Activity Analysis:

- The distribution of trades across sentiment categories revealed significant differences in trading activity.

Sentiment	Number of Trades
Fear	61,837
Greed	50,303
Extreme Greed	39,992
Neutral	37,686
Extreme Fear	21,400

- Observation:
  Fear periods recorded the highest number of trades, accounting for 61,837 transactions. In contrast, Extreme Fear periods recorded the lowest trading activity with 21,400 trades.
- Insights:

This suggests that traders remain highly active during Fear conditions, potentially due to increased market volatility and the presence of buying and selling opportunities. However, during Extreme Fear, participation decreases as uncertainty and risk perception become significantly higher.

II. Profitability Analysis:

- Average Closed Profit and Loss (PnL) was calculated to evaluate profitability under different market sentiments.

Sentiment	Average Closed PnL
Extreme Greed	67.89
Fear	54.29
Greed	42.74
Extreme Fear	34.54
Neutral	34.31

- Observation:
  Extreme Greed produced the highest average profit per trade, while Neutral sentiment produced the lowest average profitability.
- Insight:
  Highly bullish market conditions appear to create more favorable trading opportunities. Traders achieved greater profits during periods of strong market optimism compared to neutral or highly pessimistic conditions.

III. Total Profitability Analysis:

- Total realized profit was also analyzed across sentiment categories.

Sentiment	Total Profit Generated
Fear	Highest
Extreme Greed	Second Highest
Greed	Third Highest
Neutral	Lower
Extreme Fear	Lowest

- Observation:
  Fear generated the highest total profit despite not having the highest average profit per trade.
- Insight:
  The large number of trades executed during Fear periods contributed to the highest overall profitability. This indicates that trading volume played a major role in determining total profit generation.

IV. Win Rate Analysis:

- Win rate was calculated as the percentage of profitable trades.

Sentiment	Win Rate (%)
Extreme Greed	46.49
Fear	42.08
Neutral	39.70
Greed	38.48
Extreme Fear	37.06

- Observation:
  Extreme Greed achieved the highest win rate, while Extreme Fear recorded the lowest.
- Insight:
  Traders were more successful during highly optimistic market conditions and less successful during periods of extreme pessimism. This indicates that market sentiment has a measurable impact on trading outcomes.

V. Trading Volume Analysis:

- Trading volume was measured using the total Size USD traded under each sentiment.

Sentiment	Trading Volume (USD)
Fear	483.3 Million
Greed	288.6 Million
Neutral	180.2 Million
Extreme Greed	124.5 Million
Extreme Fear	114.5 Million

- Observation:
  Fear periods attracted the highest amount of trading capital.
- Insight:
  Traders allocated significantly more capital during Fear conditions, suggesting that market uncertainty and volatility encouraged greater market participation and position-taking.

VI. Long vs Short Strategy Performance:

- The profitability of long and short positions was compared across sentiment categories.
  Findings:
  - During Fear and Extreme Fear periods, short positions generated higher average profits than long positions.
  - During Greed and Extreme Greed periods, long positions generated higher average profits than short positions.
- Insight:
  The effectiveness of trading strategies appears to depend heavily on market sentiment. Bearish strategies performed better during fearful markets, while bullish strategies performed better during optimistic

markets. This finding demonstrates a clear relationship between market sentiment and optimal trading direction.

VII. Coin Preference Analysis:

- The most frequently traded cryptocurrency under each sentiment category was identified.

Sentiment	Most Traded Coin
Extreme Fear	HYPE
Fear	HYPE
Neutral	HYPE
Greed	@107
Extreme Greed	@107

- Observation:
  HYPE dominated trading activity during Fear, Extreme Fear, and Neutral market conditions. In contrast, @107 became the most actively traded asset during Greed and Extreme Greed periods.
- Insight:
  Trader preferences shifted with changing market sentiment. Different assets attracted attention under bearish and bullish market environments, indicating that sentiment may influence asset selection decisions.

8. Recommendations –

Based on the findings obtained from the analysis, the following recommendations can be considered:

I. Adapt Trading Strategy to Market Sentiment
The results indicate that market sentiment significantly influences trading performance.

- Long positions performed better during Greed and Extreme Greed conditions.
- Short positions performed better during Fear and Extreme Fear conditions.

Traders may benefit from adjusting their trading direction according to prevailing market sentiment rather than applying the same strategy across all market conditions.

II. Monitor the Fear & Greed Index as a Decision-Making Tool
The Fear & Greed Index provides valuable information regarding overall market psychology.
Since profitability and win rates varied across sentiment categories, traders can use sentiment indicators as an additional input when making trading decisions and managing risk.

III. Apply Stronger Risk Management During Extreme Fear

Extreme Fear produced the lowest win rate and relatively low average profitability.
This suggests that market conditions become more challenging during periods of intense pessimism. Traders should consider:

- Reducing position sizes
- Using tighter stop-loss levels
- Avoiding excessive leverage

during such conditions.

IV. Capitalize on Opportunities During Fear Periods
Fear periods generated the highest trading volume and the highest total profit.
This indicates that market volatility during fearful conditions may create profitable opportunities for disciplined traders who are able to manage risk effectively.

V. Track Asset Preference Changes
The analysis showed that trader preference shifted between HYPE and @107 under different sentiment conditions.
Monitoring changes in asset popularity may help traders identify emerging opportunities and better understand market behavior.

9. Conclusion –

This project investigated the relationship between Bitcoin market sentiment and trader performance using the Fear & Greed Index and historical trading data.

The analysis revealed a clear connection between market sentiment and trading outcomes. Fear periods attracted the highest trading activity and trading volume, while Extreme Greed generated the highest average profit per trade and the highest win rate.

The study also showed that trading strategy effectiveness varied across market conditions. Short positions performed better during Fear and Extreme Fear periods, whereas long positions performed better during Greed and Extreme Greed periods. Additionally, trader preferences for specific cryptocurrencies changed depending on market sentiment.

Overall, the findings suggest that market sentiment can serve as a valuable indicator for understanding trader behavior and improving decision-making. Incorporating sentiment analysis into trading strategies may help traders enhance profitability, improve risk management, and adapt more effectively to changing market conditions.